CONTROLLER / CLOCK / DEBUG

U4A

PC1 / pin 10: FAN_PWM = LPTIM1_OUT (active high; reset off).
PC13: EXT_POWER_FAULT_N = sensor U604 OR fan U503 (active low).

CAN
FDCAN1_TX — 48 — PA12
FDCAN1_RX — 47 — PA11
CAN_STB — 28 — PB10

UART
USB_UART_TX — 45 — PA9
USB_UART_RX — 46 — PA10

SPI3 / I2C2
SPI3_SCK_MCU — 54 — PB3
SPI3_MOSI_MCU — 56 — PB5
SPI3_MISO — 55 — PB4
AS5047P_CS_N — 51 — PA15
I2C2_SCL — 21 — PC4

FEEDBACK
FEEDBACK_A_H1 — 57 — PB6
FEEDBACK_B_H2 — 58 — PB7
FEEDBACK_I_H3 — 59 — PB8/BOOT0
EXT_FEEDBACK_SELECT — 53 — PD2
INPUT_MUX_ENABLE_N — 4 — PC14

IMU IRQ
IMU_INT1 — 17 — PA4
IMU_INT2 — 18 — PA5

BRAKE
BRAKE_PWM — 60 — PB9

TP1023 EXT_POWER_FAULT_N
EXT_POWER_FAULT_N — 3 — PC13

TP1021 CAN_TERM_DISABLE
CAN_TERM_DISABLE — 5 — PC15

TP1024 I2C2_SDA
I2C2_SDA — 44 — PA8

PF0 — 6
PF1 — 7
PC0 — 9 — VBUS_SENSE
PC1 — 10 — FAN_PWM
PC2 — 11 — NTC_PHASE_B
PC3 — 12 — NTC_PHASE_A
PA0 — 13 — NTC_PHASE_C
PG10/NRST — 8 — NRST
PA13/SWDIO — 49 — TP1008 SWDIO
PA14/SWCLK — 50 — TP1009 SWCLK

STSPIN32G4

Y301 24MHz 7pF
24 MHz / 7 pF load
10 pF C0G starting caps
Case pads 2 and 4 = GND

C308 10pF C0G
C309 10pF C0G

TP303 NRST
R303 2k2
3V3
SW301 RESET
C310 100nF

POWER / CAN TERMINATION INDICATOR

5V_SYS
D301 LTST-C19HE1WT
R304 15k
CAN_TERM_GATE
Q301 CSD17483F4
R315 10k

J1001 TC2030-NL SWD
VCC, RESET, SWCLK, SWDIO, SWO, GND
NRST, SWCLK, SWDIO

TC2030-NL: 6 pogo contacts, 3 locating holes.
VTREF is sense-only; power board separately.
Pin 6 (SWO) is NC; PB3 is encoder SPI3 SCK.
Keep probe approach and locating holes clear.

BLUE = 5 V power present; PURPLE = power + termination enabled.
Green unused. PC15 HIGH disables termination and red.
Reset Hi-Z defaults to PURPLE once 10 V VCC is present.
PC15: low-speed push-pull; keep LSE disabled. No status blinking.
Q301 isolates LED current from the CAN termination gate drive.
With USB-only power / no 10 V rail: blue only; termination unavailable.

U4B

MCU / ANALOG SUPPLIES

C301 100nF
C302 10uF 10V
C307 100nF

REG3V3/VDD — VBAT — REGIN — VDDA — VREF+ — VSS/EP

R301 0R
R302 0R
C303 100nF
C304 1uF
C305 100nF
C306 1uF

U4 units: A digital, B supplies, C gate driver, D/E/F current amplifiers. Same STSPIN32G4 package.

VDDA VREF+

STSPIN32G4

PC15 drives CAN termination; RGB power / termination indication

ShiroFOC

Sheet: /MCU / clock / SWD/
File: MCUclockSWD.kicad_sch

Title: Controller, clock, reset and debug

Size: A3	Date: 2026-10-06	**Rev: A3-CAN-Term**
KiCad E.D.A. 9.0.4		Id: 4/12

BRAKE CHOPPER — TWO COMMAND SOURCES, ONE GATE DRIVE

FIRMWARE / HARDWARE DIODE OR

BRAKE_PWM

R502
33R

D502
BAT54CM

BRAKE_PWM_DRV

R501
10k

GND

R505
10k

GND

LOCAL GATE DRIVE / POWER SWITCH

VCC

C501
1uF 25V

C502
100nF 50V

GND

U501
UCC27517ADBVR

VDD
IN+
IN−
GND
OUT

GND

R503
4.7R

TP501
BRAKE_GATE

Q501
ISZ080N10NM6 100V

D501
BZT52C12 12V

R504
10k

GND

VM

J501

EXTERNAL
10R / 250W*

BRAKE RESISTOR

TP502
BRAKE_SWITCH

Short / twisted resistor leads.

Validate Q501 drain overshoot.

Q501: D = pads 5–8; S = pads 1–3; G = pad 4.

CAPACITOR UN-EXPLODER

Board-bus overvoltage backup — NOT a BMS or battery protector.

VM

R506
470k 1% 75V

R509
13.3k 1%

C504
1nF C0G

R510
1M 1%

3V3

C503
100nF 50V

GND

U502
TLV3012BIDCKR

VDD
IN+
IN−
OUT
REF
GND

BRAKE_OV_CMD

The positive-feedback resistor is wired back to IN+.

Internal 1.242 V reference returns directly to IN−.

U502: nominal ON 45.8 V / OFF 44.1 V. Board-bus backup only; thresholds have tolerance and dynamic overshoot.

*Provisional external resistor: RH25010R00FE01, 10R / 250W WITH heatsink. About 212W at 46V; size for stopping energy.

Place U501 decoupling / R503 / D501 at Q501. Connect U501 pin 2 (GND) and pin 4 (IN−) to GND; Kelvin return to Q501 source.

Brake NTC removed. Brake PWM and hardware bus-overvoltage clamp remain active.

PROTECTED 5V FAN / PC1 HARDWARE PWM

5V_SYS

U503
TPS2553DBVR

IN
EN
ILIM
OUT
FAULT
GND

EXT_POWER_FAULT_N

C505
100nF 16V X7R

R513
49.9k 1%

FAN_5V

C506
4.7uF 16V X7R

D506
PMEG2010EA

J502
5V FAN

FAN_RETURN

FAN_PWM

R511
100R

FAN_GATE

Q502
CSD17483F4

R512
100k

GND

J502: 1 = FAN_5V, 2 = switched return. Do not ground pin 2.
PC1 / LPTIM1_OUT: HIGH = ON; LOW/reset = OFF. Fan target <=250mA.
Choose a fan rated for supply PWM; tune frequency/minimum duty and startup kick.
U503 limit: 520mA nominal. USB-only fan operation requires a verified source budget.

Same A0 circuit; connected functional drawing

ShiroFOC

Sheet: /Brake chopper/
File: 07_Brake_Chopper.kicad_sch

Title: Brake chopper / overvoltage protection

Size: A3	Date: 2026-09-16	**Rev: A1-Drawing**
KiCad E.D.A. 9.0.4		Id: 8/12

EXTERNAL HALL / ABI: 3.3V or 5V logic -> 3.3V Schmitt buffer -> reset isolation

External sensor supply: JP601 selects 3V3 (default) or 5V.
J601.1 = protected output; J601.2/6 = GND. No power input.
Change voltage with power off: remove 1-2 bridge before bridging 2-3.
Target sensor + pull-ups <=25mA. Fan has a separate branch on J502.

D601 rail isolated from supplies.
C604 is required; no 3V3 clamp tie.

INPUT OPTIONS
Default: 3.3V / 5V push-pull; pull-ups DNP.
Open-drain: fit R608-R610 (4.7k).
Pull-ups follow JP601 selected sensor voltage.
Do not fit for externally driven push-pull.
100k pull-downs define disconnected inputs.
Single-ended only; common ground required.

D601
TPD3E001DRYR
NC
GND
VCC
IO3
IO2
IO1
FB_ESD_RAIL
C604
100nF 50V

J601
MP
MOUNT
EXT ENCODER / HALL

JP601
3V3
5V_SYS
EXT_SUPPLY_SELECTED
SENSOR SUPPLY

U604
TPS2553DBVR
IN
EN
ILIM
OUT
FAULT
GND
EXT_VCC
EXT_POWER_FAULT_N
C606
4.7uF 16V X7R

R604
10k

U604: ILIM tied to IN => 75mA nominal (50-100mA range).
EN follows selected input: continuous sensor power.
EXT_POWER_FAULT_N -> PC13; wired-OR with fan limiter U503.
Stop motor if external feedback is lost; fault does not identify branch.

R608
4.7k DNP
R609
4.7k DNP
R610
4.7k DNP

R605
100R
R606
100R
R607
100R

U603A
U603C
U603B
74LVC3G17

R615
100k
R616
100k
R617
100k

U602A
TMUX1574PW
S1A
S1B
S2A
S2B
S3A
S3B
D1
D2
D3

TP1015
FEEDBACK_A_H1
TP1016
FEEDBACK_B_H2
FEEDBACK_J_H3

R307
10k

C607
100nF 16V X7R

U603D
74LVC3G17

C605
100nF

R305
10k
R306
10k

INPUT_MUX_ENABLE_N
EXT_FEEDBACK_SELECT

U602C
TMUX1574PW
VDD
SEL
EN_N

C603
100nF

U602B
TMUX1574PW
VBUS_MUX_IN
VBUS_SENSE

Reset: mux disabled. PB8/BOOT0 held low by R307.
EN_N=0 enables VM sensing; wait >=2ms before ADC read.
SEL=0 grounds TIM4 inputs; SEL=1 connects external Hall/ABI.
Onboard absolute-angle SPI is independent of this switch.

3.3V / 5V external inputs; interface revision
ShiroFOC
Sheet: /External feedback/
File: 09_Feedback_Selection.kicad_sch
Title: External Hall / ABI conditioning and reset isolation
Size: A3 | Date: 2026-09-26 | **Rev: A1-Drawing**
KiCad E.D.A. 9.0.4 | Id: 10/12